USN | | | | | | | | | | [23ESEC11]

FIRST SEMESTER B.TECH EXAMINATION, JANUARY - 2024

ELEMENTS OF COMPUTER ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer all Questions

		PART - A	**05 X 02=10**		
1.	**a**	Explain any two advantages and disadvantages of second generation computer.	2	CO1	**02**
	b.	Write short note on manufacturer–programmed read only memory.	2	CO2	**02**
	c.	What is application software and write any two examples.	1	CO3	**02**
	d.	Write a C Program to Print an Integer entered by the user.	3	CO5	**02**
	e.	Mention different decision making statements supported by C.	1	CO5	**02**
		PART - B	**06 X 05=30**		
2.	**a.**	Compare first generation computers with second generation computers.	2	CO1	**05**
		OR			
	b.	Explain different locator devices in computer system.	2	CO1	**05**
3.	**a.**	What is memory system and explain the different types of memory.	2	CO2	**05**
		OR			
	b.	Write the difference between SRAM and DRAM.	2	CO2	**05**
4.	**a.**	Explain the important features of system software.	2	CO3	**05**
		OR			
	b.	Explain the benefits of communications software.	2	CO3	**05**
5.	**a.**	Explain implicit and explicit type of conversion with example.	2	CO4	**05**
		OR			
	b.	Write a C-program to display 'JAIN University' 10 times using 'for' loop.	3	CO4	**05**
6.	**a.**	Explain 10 keywords supported in C with their function.	2	CO4	**05**
		OR			
	b.	Write a C-program to find sum and average of 10 numbers.	3	CO6	**05**

7.	**a.**	Explain the basic concept of decision making and explain the working of 'if' statement.	2	CO5	**05**
		OR			
	b.	Write a C program to print the Fibonacci series up to 10 numbers.	3	CO6	**05**
		PART - C	**06 X 10=60**		
8.	**a.**	Explain the working of computer system with the help of a block diagram.	2	CO1	**10**
		OR			
	b.	Briefly explain the five generations of computers.	2	CO1	**10**
9.	**a.**	What is primary memory? Explain the different types read only memory.	2	CO2	**10**
		OR			
	b.	Explain the structure of hard disk and briefly explain the mechanism of read/write operation.	2	CO2	**10**
10.	**a.**	Define software and explain the software hierarchy.	2	CO3	**10**
		OR			
	b.	Explain with neat block diagram of assembler, compiler and interpreter.	2	CO3	**10**
11.	**a.**	Explain the structure of 'C' program syntax with an example.	3	CO4	**10**
		OR			
	b.	Explain the different logical operators supported in C and write a C-program to find the largest number among three numbers.	3	CO4	**10**
12.	**a.**	Write syntax of scanf and printf with example and write commonly used scanf and printf format codes.	2	CO5	**10**
		OR			
	b.	Explain the syntax of if-else statement and write a C program to find entered number is positive or negative.	3	CO6	**10**
13.	**a.**	Give the syntax of while statement. Write a C program to display N natural numbers.	3	CO6	**10**
		OR			
	b.	Write a C Program to display prime numbers between 50 and 100.	3	CO6	**10**